Optical Flow Matching: Reframing Optical Flow as Continuous Transport Dynamics

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Abstract

Modern optical flow estimation, though empowered by recent deep neural architectures, remains rooted in the discrete correspondence paradigm inherited from classical vision. Most networks infer frame-to-frame displacements, capturing where pixels move but not how motion evolves continuously through time. Yet physical motion in the real world follows smooth dynamics governed by underlying velocity fields, as long established in fluid mechanics and transport theory. To bridge this gap, we introduce Optical Flow Matching (OFM), a continuous formulation that learns a time-dependent velocity field to transport pixel coordinates along motion distribution coherent trajectories. A key component of our OFM is Triangle Velocities Synergy (TVS), a neat yet pragmatic geometric mechanism that provides a stable and physically meaningful velocity construction, ensuring that continuous transport remains well-defined. Combined with an Euler-based ODE solver, OFM yields flow fields that are temporally smooth, geometrically consistent, and process-interpretable. Experiments on Sintel, KITTI, and Spring demonstrate that OFM achieves state-of-the-art accuracy, enhanced temporal stability, and notably stronger cross-dataset generalization, advancing optical flow estimation from correspondence inference to continuous dynamical reasoning. Code is available at `https://github.com/LA30/OFM`.

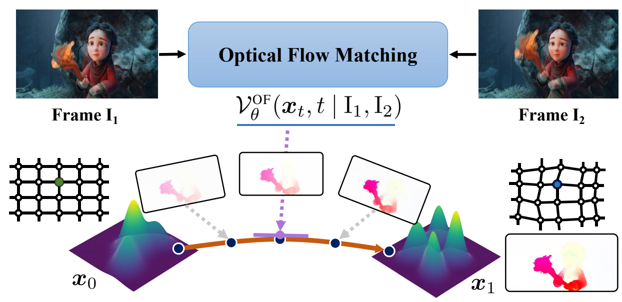

Figure 1. **Idea Illustration.** Optical Flow Matching (OFM) reframes optical flow as a continuous transport process, where a learned velocity field $\mathcal{V}_\theta^{\mathrm{OF}}$ evolves pixel coordinates $(x_0 \rightarrow x_1)$ continuously along time t, moving beyond discrete displacement estimation. Best viewed in color.

1. Introduction

Understanding motion lies at the core of visual perception, shaping how both biological and artificial systems interpret and anticipate dynamic scenes. Optical flow, the dense field that describes how pixels move between frames, serves as a fundamental representation of scene dynamics. Over the past decades, it has driven a wide range of vision tasks, including video analysis, 3D and scene-flow reconstruction, and motion segmentation [3, 16, 35, 37, 40], as well as emerging areas such as video generation, video frame interpolation, and video inpainting [22–24]. Acting as a bridge between perception and dynamics, optical flow not only enables motion interpretation but increasingly serves as a prior for motion synthesis and control.

Despite its central role, the conceptual foundation of optical flow has changed little since its inception. Classical methods formulate flow estimation as a correspondence problem, recovering pixel displacements between adjacent frames under brightness constancy and spatial smoothness assumptions. Modern state-of-the-art architectures such as PWC-Net [46], RAFT [51], and GMFlow [57] have substantially improved accuracy, yet they remain bound to this discrete paradigm, interpreting motion as pairwise feature alignment. While such models excel at estimating displacements, they detach motion from the underlying physical process that generates it. Consequently, these models recover the outcomes of motion, *i.e.,* pixel displacements, without learning the dynamics that give rise to them.

To transcend this limitation, we reformulate optical flow as a problem of continuous transport dynamics rather than discrete displacement regression. Drawing inspiration from Flow Matching [25], we reinterpret optical flow estimation as learning a time-dependent velocity field that continuously transports visual representations from one frame to

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another. Instead of predicting isolated displacements, our model learns the governing field that describes how each pixel's position evolves smoothly over time. Modeling motion in this continuous manner restores the physical relationship between velocity and displacement, enabling the network to reason about how motion develops rather than merely where it occurs. We term this formulation Optical Flow Matching (OFM), to our knowledge, the first approach in the field of optical flow that bridges generative transport theory with visual motion estimation.

Practically, OFM is built on conditional optimal transport and an Euler-based ODE solver that evolves pixel trajectories continuously over time. In Flow Matching [25], trajectories are initiated from Gaussian noise and evolve through a learned velocity field that transports samples from noise to data in a probabilistic space. In contrast, we introduce Triangle Velocities Synergy to help adjust the objective of OFM, namely OFM-TVS, which effectively reconciles the theoretical guarantees of Flow Matching with the well-established training objectives of optical flow models. The core procedures include introducing auxiliary trajectories and performing a triangular geometric conversion for the velocity vectors. As OFM-TVS is a coordinate-based formulation that grounds Flow Matching's transport behavior, it cannot follow image generation tasks and simply sample from Gaussian noise initially. Instead, we define that each coordinate starts from a locally perturbed position, obtained by adding a small Gaussian offset to its original location. This perturbed coordinate acts as a physically meaningful analogue of Flow Matching's noisy initialization, enabling the model to reason about motion in tangible image geometry rather than in an abstract latent space.

The proposed approach transfers the continuous transport principles of Flow Matching into the domain of physical motion (see Fig. 1), producing flow fields that remain temporally stable, geometrically consistent, and process-interpretable. Extensive experiments demonstrate that our OFM consistently surpasses the recent displacement-based methods on challenging optical flow benchmarks. Overall, the **contributions** of this work can be summarized as:

- **Redefining the Paradigm.** We reformulate optical flow estimation from discrete correspondence inference to continuous transport dynamics, establishing a physically grounded view of motion as time-evolving velocity fields.
- **Methodological Innovation.** We propose Optical Flow Matching with Triangle Velocities Synergy (OFM-TVS), an approach built on the classic conditional optimal transport and an Euler-based ODE solver, and enhanced with an ingenious geometric transformation, which effectively reconciles the theoretical guarantees of Flow Matching with a practically well-grounded training objective.
- **SOTA Performance.** OFM can be seamlessly integrated into various optical flow architectures, consistently improving accuracy and temporal stability on Sintel, KITTI, and Spring, and advancing optical flow estimation from static mapping to dynamical reasoning.

2. Related Works

Optical Flow. Optical flow estimation has long been framed as a discrete correspondence problem, seeking pixel-wise displacements between frames under brightness constancy and spatial smoothness assumptions [2, 12]. Modern deep baselines crystallized around hierarchical warping with cost volumes (PWC-Net) and, subsequently, dense all-pairs correlation with iterative refinement (RAFT), which set the template for two-frame estimation [46, 51]. Recent works have made improvements from various perspectives based on the RAFT model [19, 20, 30, 47, 55, 56, 59, 61]. Building on this template, recent works expand spatial range and data efficiency through global matching and transformerized cost reasoning, *e.g.*, GMFlow [57] casts flow as global feature matching, while FlowFormer [13] encodes 4D cost volumes as reusable cost memory, with FlowFormer++ [42] demonstrating masked cost-volume pretraining gains across datasets. Robustness and temporal coherence have been further enhanced by motion aggregation [18, 28, 29] and multi-frame reasoning [6, 41, 52], while semi/self-supervised methods reduce label dependency [11, 15]. Significant advances have also been made in methods for estimating optical flow on other types of data [32, 33, 58]. Generative diffusion approaches (*e.g.*, FlowDiffuser [31], DDVM [39]) introduce uncertainty modeling yet remain confined to discrete displacement distributions without explicit velocity–displacement coupling [31, 39]. In contrast, our Optical Flow Matching (OFM) learns a time-dependent velocity field and integrates it via an Euler-based ODE solver, explicitly linking pixel displacements to their underlying motion dynamics for temporally coherent and physically consistent flow.

Flow Matching and Generative Transport. Recent advances in generative transport reformulate data generation as continuous-time dynamics governed by an ODE. Score-based diffusion models established the probability–flow perspective linking diffusion and deterministic sampling [43]. Flow Matching [25] further simplified training by directly learning a velocity field along user-defined transport paths, while Rectified Flow [27] stabilized convergence through straight-line paths and consistent supervision. Building on these foundations, Consistency Models [44] and MeanFlow [10] improved sample efficiency via single-step or amortized integration, making continuous transport practical for high-dimensional vision tasks. In this paper, our OFM extends Flow Matching theory to the optical flow domain, establishing a new formulation that models pixel motion as continuous coordinate transport governed by the marginal velocity and an ODE solver.

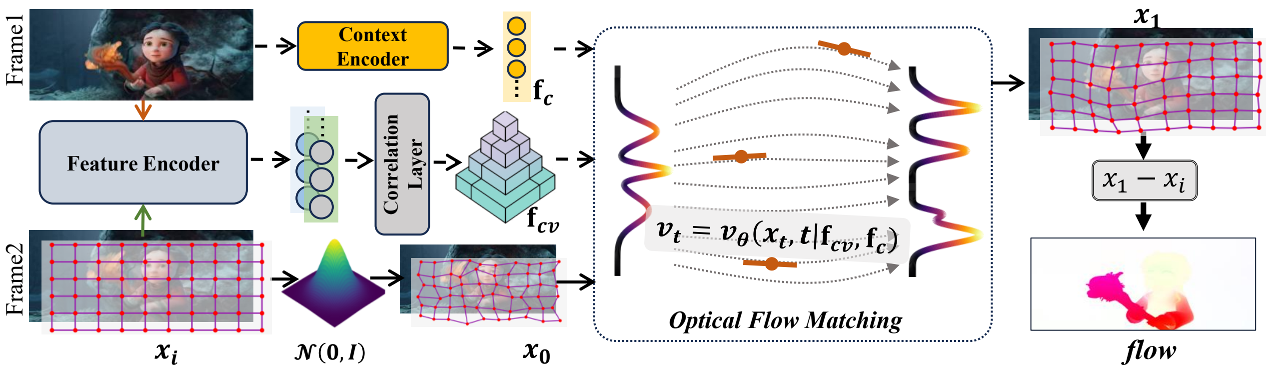


Figure 2. **Overview of Optical Flow Matching (OFM).** Given an image pair, the network extracts context features and a correlation volume to parameterize a time-dependent velocity field. Starting from perturbed pixel coordinates $\boldsymbol{x}_0$ (around initial coordinate $\boldsymbol{x}_i$), OFM integrates this field to evolve coordinates $\boldsymbol{x}_t$ toward their final positions $\boldsymbol{x}_1$. The optical flow is obtained as the displacement $\boldsymbol{x}_1 - \boldsymbol{x}_i$.

3. Methodology

Our Optical Flow Matching bridges the theoretical principles of flow matching with the practical challenges of optical flow estimation. As illustrated in Fig. 2, it operationalizes a continuous probabilistic formulation within a RAFT-like architecture, thereby linking abstract flow dynamics to concrete optical flow estimation. The following sections introduce the theoretical preliminaries of flow matching (SEC. 3.1), extend the formulation to optical flow estimation as detailed in SEC. 3.2, and conclude with its architectural instantiation in SEC. 3.3.

3.1. Preliminaries on Flow Matching

Among numerous generative paradigms, flow-based models have emerged as a powerful framework [25], characterized by their ability to construct a direct, continuous mapping formulated through an Ordinary Differential Equation (ODE). The central objective is to learn a velocity field, denoted as $\boldsymbol{v}_t(\boldsymbol{x}_t)$, which defines the dynamics of transforming a sample from a simple prior distribution $p_0(\boldsymbol{\epsilon})$, *e.g.*, Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, to a sample from the complex data distribution $p_1(\boldsymbol{x})$. This formulation enables the construction of a probability path such that

$$\boldsymbol{x}_t = a_t \boldsymbol{x} + b_t \boldsymbol{\epsilon}, \tag{1}$$

with time t, in which a_t and b_t represent predefined schedules. The velocity $\boldsymbol{v}_t$ is specified as $\boldsymbol{v}_t = d\boldsymbol{x}_t/dt$, the time derivative of $\boldsymbol{x}_t$. As established in [25], this is the *conditional* velocity, written as $\boldsymbol{v}_t = \boldsymbol{v}(\boldsymbol{x}_t \mid \boldsymbol{x})$.

Because multiple latent pairs $(\boldsymbol{x}, \boldsymbol{\epsilon})$ can generate the same $\boldsymbol{x}_t$ and corresponding $\boldsymbol{v}_t$, the Flow Matching objective considers the expected value over these latent pairs, producing the *marginal* velocity [25]:

$$\boldsymbol{v}(\boldsymbol{x}_t, t) \triangleq \mathbb{E}_{p_t(\boldsymbol{v}_t|\boldsymbol{x}_t)}[\boldsymbol{v}_t]. \tag{2}$$

A neural network $\boldsymbol{v}_\theta$, parameterized by θ, is optimized to fit the marginal velocity field:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,p_t(\boldsymbol{x}_t)} \|\boldsymbol{v}_\theta(\boldsymbol{x}_t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)\|^2. \tag{3}$$

With the estimated marginal velocity $v(z_t, t)$, samples can be generated by integrating the corresponding ODE for $\boldsymbol{x}_t$:

$$\frac{d\boldsymbol{x}_t}{dt} = \boldsymbol{v}(\boldsymbol{x}_t, t). \tag{4}$$

In practice, numerical approximation is required, typically via discretization in time. Approaches such as the Euler method or other higher-order ODE solvers can be adopted.

3.2. Optical Flow Matching

3.2.1. Formulation

Conventional optical flow estimation remains rooted in a discrete formulation, where motion is inferred as pairwise feature alignment between consecutive frames. While correlation and cost-volume architectures have achieved remarkable progress, these models still tend to regress displacements rather than explicitly represent the underlying motion dynamics, which may reduce temporal consistency under occlusion, large motion, or illumination changes.

To overcome these limitations, we propose Optical Flow Matching (OFM), a continuous formulation that learns a time-dependent velocity field to transport pixel coordinates in the image domain. Instead of predicting static displacements, OFM models how coordinates evolve continuously over time under an implicit motion field. In the context of flow matching, the probability trajectory in Eq. (1) is reformulated as

$$\boldsymbol{x}_t = a_t \boldsymbol{x}_1 + b_t \boldsymbol{x}_0, \tag{5}$$

where $\boldsymbol{x}_0$ is sampled from a prior coordinate distribution, and $\boldsymbol{x}_1$ denotes the target coordinate distribution derived from ground-truth optical flow. The optical flow $\boldsymbol{f}$ specifies pixel displacements from I_1 to I_2. Thus, given images $(\mathrm{I}_1, \mathrm{I}_2)$ and pixel coordinate $\boldsymbol{x}_0$ in I_1, predicting $\boldsymbol{f}$ amounts to predicting $\boldsymbol{x}_1$ in I_2, *i.e.*, post-motion pixel positions. In addition, a scaled Gaussian perturbation is added to initial coordinates to obtain relative coordinates aligned with Flow Matching's noisy initialization, expressed as $\boldsymbol{x}_0 = \boldsymbol{x}_i + \alpha\boldsymbol{\epsilon}$, where $\boldsymbol{x}_i$ is the initial coordinate, α is a scaling factor, and $\boldsymbol{\epsilon}$ follows a standard Gaussian distribution.

The objective of Flow Matching is to learn a velocity field network that characterizes the instantaneous motion of samples along a continuous trajectory. In our formulation, optical flow networks are adopted to predict $\mathcal{V}_\theta^{\text{OF}}(\boldsymbol{x}_t, t \mid \mathrm{I}_1, \mathrm{I}_2)$, with the marginal velocity in Eq. (2) serving as the objective. Since evaluating the marginal loss directly is impractical due to the required marginalization, we follow prior work and apply the *conditional* Flow Matching loss [25]:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,\boldsymbol{x}_1,\boldsymbol{x}_0} \|\mathcal{V}_\theta^{\text{OF}}(\boldsymbol{x}_t, t \mid \mathrm{I}_1, \mathrm{I}_2) - \boldsymbol{v}_t(\boldsymbol{x}_t \mid \boldsymbol{x}_1)\|^2 \tag{6}$$

where the target $\boldsymbol{v}_t$ denotes the conditional velocity. As discussed in [25], minimization of $\mathcal{L}_{\text{CFM}}$ is theoretically equivalent to minimizing $\mathcal{L}_{\text{FM}}$.

Optimal Transport. To validate the fundamental capability of Flow Matching, we employ the classical formulation of *conditional optimal-transport* [26]. In particular, the linear conditional optimal-transport path is expressed as

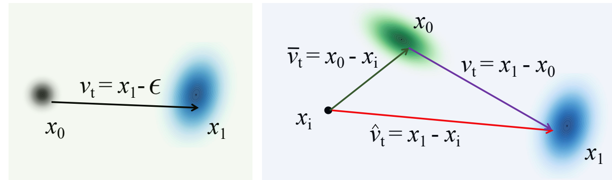
$$\boldsymbol{x}_t = t\boldsymbol{x}_1 + (1-t)\boldsymbol{x}_0, \tag{7}$$

which results in $\boldsymbol{v}_t(\boldsymbol{x}_t \mid \boldsymbol{x}_1) = \boldsymbol{x}_1 - \boldsymbol{x}_0$, making the loss function in Eq. (6) feasible.

3.2.2. Triangle Velocities Synergy

The formulation presented in Eq. (7), which we refer to as *OFM-Naive*, is theoretically well-founded but proves problematic in practice. The core issue lies in the misalignment between the theoretical objective of Flow Matching and the practical goal of optical flow estimation. From a theoretical standpoint, an optical flow model should estimate $\boldsymbol{x}_1 - \boldsymbol{x}_0$, that is, $\boldsymbol{x}_1 - (\boldsymbol{x}_i + \alpha\boldsymbol{\epsilon})$, equivalent to $\boldsymbol{f} - \alpha\boldsymbol{\epsilon}$. However, conventional optical flow networks are explicitly designed with a well-defined physical objective, *i.e.*, to establish pixel-to-pixel correspondences between adjacent frames by leveraging feature correlations via cost volumes or Transformer-based attention mechanisms. When such architectures are repurposed to predict an abstract quantity like "flow minus Gaussian noise", which lacks clear physical meaning or geometric grounding, convergence becomes highly unstable and training performance degrades substantially.

Conversely, if we prioritize the model's design, we may set $\alpha = 0$, which reduces the prior distribution $p_0(\boldsymbol{x}_0)$ to a Dirac delta function, *i.e.*, a constant $\boldsymbol{x}_0$. However, this simplification introduces another issue: it removes the diversity of conditional trajectories. In Flow Matching, the variation among conditional probability paths $p_t(\boldsymbol{x} \mid \boldsymbol{x}_1)$ arises from the stochasticity of the starting point $\boldsymbol{x}_0$. When $\boldsymbol{x}_0$ becomes constant, all trajectories originate from the same point, leading to a degenerate training objective. As a result, the model is forced to solve a more challenging mathematical problem, *i.e.*, learning a velocity field capable of transporting a single initial state to the entire complex target distribution.



(a) OFM-Naive (b) OFM-TVS

Figure 3. **Difference between OFM-Naive (a) and OFM-TVS (b).** OFM-TVS replaces the single perturbed velocity in OFM-Naive with a geometrically consistent triangular decomposition.

To address this dilemma, we propose Triangle Velocities Synergy (TVS) strategy, which effectively reconciles the theoretical guarantees of Flow Matching with the well-established training objectives of optical flow models. Specifically, we redefine the starting point as $\boldsymbol{x}_0 = \boldsymbol{x}_l + \alpha\boldsymbol{\epsilon}$, where $\boldsymbol{x}_l$ is a newly introduced constant reference. Compared to the *OFM-Naive* formulation, this modification shifts the center of the $\boldsymbol{x}_0$ distribution from $\boldsymbol{x}_i$ to $\boldsymbol{x}_l$. Furthermore, we introduce two additional probability paths associated with the original trajectory from $\boldsymbol{x}_0$ to $\boldsymbol{x}_1$. Following the linear interpolation model described in Eq. (7), these two auxiliary paths are defined as

$$\begin{aligned} \boldsymbol{y}_t &= t\boldsymbol{x}_1 + (1-t)\boldsymbol{x}_i, \\ \boldsymbol{z}_t &= t\boldsymbol{x}_0 + (1-t)\boldsymbol{x}_i, \end{aligned} \tag{8}$$

all utilizing the same time parameter t. Building upon this, the other two marginal velocities are defined as $\boldsymbol{v}_t(\boldsymbol{y}_t \mid \boldsymbol{x}_1) = \boldsymbol{x}_1 - \boldsymbol{x}_i$ and $\boldsymbol{v}_t(\boldsymbol{z}_t \mid \boldsymbol{x}_0) = \boldsymbol{x}_0 - \boldsymbol{x}_i$.

A key insight here is that, although the velocity field in Flow Matching typically represents an instantaneous quantity, the specific formulation based on conditional optimal-transport paths leads to each marginal velocity being constant for a given data sample $\boldsymbol{x}_1$. Geometrically, these three constant vectors form a triangle, as illustrated in Fig. 3. Consequently, this formulation establishes a rigorous and practical bridge between the theoretical objective of Flow Matching, characterized by $\boldsymbol{v}_t(\boldsymbol{x}_t \mid \boldsymbol{x}_1)$, and the tangible training signal $\boldsymbol{v}_t(\boldsymbol{y}_t \mid \boldsymbol{x}_1)$ employed by optical flow models, thereby making the overall implementation feasible and stable in practice.

The configuration of $\boldsymbol{x}_l$ requires careful consideration, as it significantly influences estimation performance. Optical flow estimation generally incurs higher errors for large displacements compared to small ones. Ideally, $\boldsymbol{x}_l$ should be placed near the target $\boldsymbol{x}_1$ to minimize error. However, since the target distribution's direction is unknown beforehand, a fixed constant cannot universally suit all cases. To address this, we treat $\boldsymbol{x}_l$ as a learnable variable to be predicted by the model, *i.e.*, the prediction of coarse global flow, which im-

Algorithm 1 OFM-TVS Training Procedure

Input: Image pair $(\mathrm{I}_1, \mathrm{I}_2)$ with ground-truth flow $\boldsymbol{f}_{gt}$, initial coordinate $\boldsymbol{x}_i$, and total training iterations N.
Output: Optimized OFM generator $\mathcal{V}_\theta^{\text{OF}}$.

1: **for** $i = 1$ to N **do**
2: $\quad \boldsymbol{x}_l \leftarrow \mathcal{V}_\theta^{\text{OF}}(\mathrm{I}_1, \mathrm{I}_2)$
3: $\quad p_l(\boldsymbol{x}) \leftarrow \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{x}_l, \boldsymbol{I})$
4: $\quad$ Sample $\boldsymbol{x}_0 \sim p_l(\boldsymbol{x}), \quad t \sim \mathcal{U}(0, 1)$
5: $\quad \boldsymbol{x}_1 \leftarrow \boldsymbol{x}_i + \boldsymbol{f}_{gt}$
6: $\quad \boldsymbol{x}_t \leftarrow t\boldsymbol{x}_1 + (1-t)\boldsymbol{x}_0$
7: $\quad$ *// Triangle Velocities Synergy objective adaptation*
8: $\quad \hat{\boldsymbol{v}}_t \leftarrow \mathcal{V}_\theta^{\text{OF}}(\boldsymbol{x}_t, t \mid \mathrm{I}_1, \mathrm{I}_2)$
9: $\quad \mathcal{L}(\theta) = \|\hat{\boldsymbol{v}}_t - \boldsymbol{f}_{gt}\|_2^2$
10: $\quad$ Update $\mathcal{V}_\theta^{\text{OF}}$ by minimizing $\mathcal{L}(\theta)$
11: **end for**

Algorithm 2 OFM-TVS Sampling Procedure

Input: Image pair $(\mathrm{I}_1, \mathrm{I}_2)$, initial coordinate $\boldsymbol{x}_i$, and total sampling iterations K.
Output: Generated sample $\boldsymbol{x}_t$.

// Solving the ODE via Euler integration
$t_\Delta \leftarrow 1/K$
$\boldsymbol{x}_l \leftarrow \mathcal{V}_\theta^{\text{OF}}(\mathrm{I}_1, \mathrm{I}_2)$
$p_l(\boldsymbol{x}) \leftarrow \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{x}_l, \boldsymbol{I})$
Sample $\boldsymbol{x}_0 \sim p_l(\boldsymbol{x})$
Initialize $\boldsymbol{x}_t \leftarrow \boldsymbol{x}_0$
for $i = 1$ to K **do**
$\quad t \leftarrow i/K$
$\quad$ *// Predicting the instantaneous velocity field*
$\quad \hat{\boldsymbol{v}}_t \leftarrow \mathcal{V}_\theta^{\text{OF}}(\boldsymbol{x}_t, t \mid \mathrm{I}_1, \mathrm{I}_2)$
$\quad \bar{\boldsymbol{v}}_t \leftarrow \boldsymbol{x}_0 - \boldsymbol{x}_i$
$\quad$ *// Computing the adjusted target velocity*
$\quad \boldsymbol{v}_t \leftarrow \hat{\boldsymbol{v}}_t - \bar{\boldsymbol{v}}_t$
$\quad \boldsymbol{x}_t \leftarrow \boldsymbol{x}_t + \boldsymbol{v}_t \cdot t_\Delta$
end for

plies that $\boldsymbol{x}_0$ follows a Gaussian mixture distribution $p_l(\boldsymbol{x})$ with components centered at $\boldsymbol{x}_l$.

Training. The training procedure of the proposed OFM-TVS is shown in Algorithm 1. The core innovation, detailed in lines 8–9, involves applying the velocity vector triangle relation from our proposed TVS strategy, defined by the equation $\boldsymbol{v}_t$ as

$$\boldsymbol{v}_t(\boldsymbol{x}_t \mid \boldsymbol{x}_1) = \boldsymbol{v}_t(\boldsymbol{y}_t \mid \boldsymbol{x}_1) - \boldsymbol{v}_t(\boldsymbol{z}_t \mid \boldsymbol{x}_0). \tag{9}$$

This strategic formulation allows the model $\mathcal{V}_\theta^{\text{OF}}$ to bypass direct estimation of the canonical Flow Matching objective $\boldsymbol{v}_t(\boldsymbol{x}_t \mid \boldsymbol{x}_1)$. Instead, the learning is channeled through a closely related proxy velocity $\hat{\boldsymbol{v}}_t = \boldsymbol{v}_t(\boldsymbol{y}_t \mid \boldsymbol{x}_1)$. A key advantage of this approach is that the proxy vector $\boldsymbol{v}_t(\boldsymbol{y}_t \mid \boldsymbol{x}_1) = \boldsymbol{x}_1 - \boldsymbol{x}_i$ inherently represents the optical flow field. Consequently, the model can be effectively supervised using standard optical flow ground truth $\boldsymbol{f}_{gt}$, ensuring seamless compatibility with the established supervised learning paradigm in optical flow estimation.

Sampling. During inference, the integration along the probability path is performed numerically by discretizing time. To simplify the sampling process, our OFM employs the Euler method for the numerical solution of the ODE. Given an estimate $\bar{\boldsymbol{v}}_t$ of $\boldsymbol{v}_t(\boldsymbol{z}_t \mid \boldsymbol{x}_0)$, the core step involves using the vector triangle relation from the proposed TVS strategy to compute $\boldsymbol{v}_t = \hat{\boldsymbol{v}}_t - \bar{\boldsymbol{v}}_t$, as presented in Eq. (9). The computed velocity $\boldsymbol{v}_t$ is then used in the ODE sampling process to estimate $\boldsymbol{x}_t$. The overall sampling algorithm is outlined in Algorithm 2.

3.3. Instantiation

The theoretical framework developed above, based on Optical Flow Matching and Triangle Velocities Synergy, functions entirely at the algorithmic level with no dependence on a specific neural parameterization. This abstraction ensures strong generality, allowing it to be seamlessly instantiated within the entire family of RAFT-like architectures, which represent the dominant class of iterative optimization frameworks in contemporary optical flow estimation [13, 17, 51, 54]. In a standard RAFT-style pipeline, context feature $\mathbf{f}_c$ and cost-volume feature $\mathbf{f}_{cv}$ are extracted and encoded from the image pair $(\mathrm{I}_1, \mathrm{I}_2)$, and a recurrent decoder takes them as inputs and iteratively refines the correlation state $coord_1$ from $coord_0$, yielding the final optical flow $\boldsymbol{f}_{\text{pred}} = coord_1 - coord_0$.

In our OFM-TVS design, the variable $\boldsymbol{x}_t$ models the probabilistic path that leads toward the ground-truth $coord_1$. The proposed model $\mathcal{V}_\theta^{\text{OF}}$ builds upon a conventional optical flow network, conditioned on the image pair $(\mathrm{I}_1, \mathrm{I}_2)$ and augmented with two additional inputs, $\boldsymbol{x}_t$ and t. Specifically, we follow prior works [13, 31, 42] to adopt Twins-SVT as the encoders, with the global flow being produced by softmax-based feature matching [57]. It is worth noting that we only employ the parameter-free global matching specifically for $\boldsymbol{x}_l$ in our TVS strategy, while omitting the flow propagation and refinement modules [57] to minimize additional computational cost. Moreover, we follow FlowDiffuser [31] to develop the decoder with its time-relevant components and hidden-state training strategy.

Unlike traditional flow decoders, $\mathcal{V}_\theta^{\text{OF}}$ takes $\boldsymbol{x}_t$ rather than $coord_1$ as its state input, while preserving the iterative refinement mechanism characteristic of RAFT-like estimation. In the decoding stage, the velocity of the probability path at time t is estimated as $\boldsymbol{v}_t = \boldsymbol{v}_\theta(\boldsymbol{x}_t, t \mid \mathbf{f}_{cv}, \mathbf{f}_c)$.

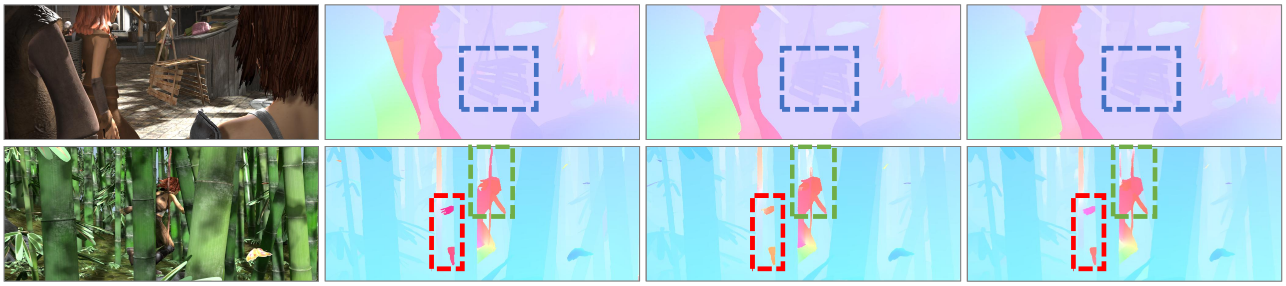


Figure 4. **Qualitative Comparison on Sintel.** OFM produces sharper boundaries and more coherent motion in challenging regions (dashed boxes), outperforming DPFlow [36] and SEA-RAFT (L) [54] in both structural fidelity and fine-grained detail.

Table 1. **Quantitative Results on Standard Benchmarks.** We compare OFM against leading two-frame methods using official metrics, and average ranks are reported for overall comparison. Best and second-best results are shown in **bold** and underline, respectively.

Method	Venue	Sintel (train)		KITTI-15 (train)		Sintel (test)		KITTI-15 (test)	Avg.
		Clean	Final	Fl-epe	Fl-all	Clean	Final	F1-all	rank
RAFT [51]	ECCV'20	1.43	2.71	5.04	17.4	1.61	2.86	5.10	13.3
GMA [17]	ICCV'21	1.30	2.74	4.69	17.1	1.39	2.47	5.15	12.0
GMFlowNet [60]	CVPR'22	1.14	2.71	4.24	15.4	1.39	2.65	4.79	10.3
GMFlow [57]	CVPR'22	1.08	2.48	7.77	23.4	1.74	2.90	9.32	13.1
CRAFT [45]	CVPR'22	1.27	2.79	4.88	17.5	1.45	2.42	4.79	11.9
SKFlow [49]	NeurIPS'22	1.22	2.46	4.27	15.5	1.28	2.27	4.84	9.3
FlowFormer [13]	ECCV'22	1.01	2.40	4.09	14.7	1.16	2.09	4.68	6.4
RAFT-it [48]	ECCV'22	1.74	2.41	4.18	13.4	1.55	2.90	4.31	9.6
FlowFormer++ [42]	CVPR'23	0.90	2.30	3.93	14.1	1.07	1.94	4.52	4.7
MatchFlow [7]	CVPR'23	1.03	2.45	4.08	15.6	1.16	2.37	4.63	7.3
EMD-L [5]	ICCV'23	0.88	2.55	4.12	13.5	1.32	2.51	4.51	7.4
FlowDiffuser [31]	CVPR'24	0.86	2.19	3.61	11.8	1.02	2.03	4.17	2.7
SEA-RAFT (L) [54]	ECCV'24	1.19	4.11	3.62	12.9	1.31	2.60	4.30	8.0
DPFlow [36]	CVPR'25	1.02	2.26	3.37	11.1	1.04	1.97	**3.56**	2.9
OFM (ours)	–	**0.81**	**2.16**	**3.32**	**10.9**	**0.94**	**1.85**	3.78	**1.1**

Under the TVS strategy, the predicted velocity $\hat{\boldsymbol{v}}_t$ is directly supervised during training (see Algorithm 1), and during inference, it is converted into $\boldsymbol{v}_t$ to perform sampling steps (see Algorithm 2). The final prediction $\boldsymbol{x}_{pred}$ is then obtained, from which the optical flow is computed as $\boldsymbol{f}_{pred} = \boldsymbol{x}_{pred} - coord_0$, maintaining consistency with existing architectures. The default setting is OFM-TVS, but we omit "TVS" for simplicity, referring to it as OFM in the section of experiments.

4. Experiments

4.1. Experimental Setup

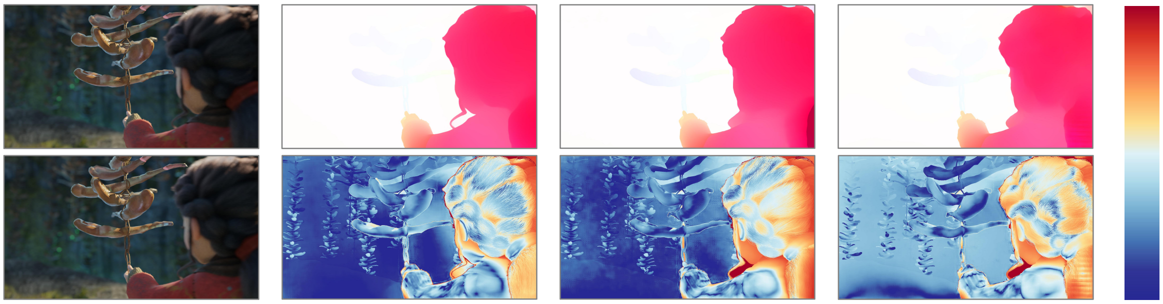
Datasets. We follow standard optical flow practice and use FlyingChairs [8], FlyingThings3D [34], Sintel [4], KITTI 2015 [35], HD1K [21], and TartanAir [53], covering both synthetic and real-world motion benchmarks.

Training Schedule. We follow the multi-stage training strategy widely adopted in recent optical flow methods [17, 49, 51]. Our training consists of four standard stages. Stage 1 trains the model on FlyingChairs for 120k iterations, and Stage 2 adds 150k iterations on FlyingThings3D, forming the widely used "C+T" schedule. Stage 3 further trains the model for 150k iterations on the mixed "C+T+S+K+H" dataset, after which the resulting model is used for online evaluation on the Sintel and Spring benchmarks. Finally, Stage 4 fine-tunes the Stage 3 model on the KITTI training split for an additional 5k iterations to generate predictions for the KITTI online benchmark. Pre-training on TartanAir [53] for extra data evaluation is also conducted, following recent works [1, 54], and is regarded as stage 0 in the training protocol.

Inference. All evaluations and submissions are conducted with a batch size of 1 on a single NVIDIA RTX 4090 GPU.

4.2. Results on Sintel and KITTI

Following prior works [31, 51], we begin by evaluating the generalization capability of OFM using the "C+T" training protocol on the Sintel [4] and KITTI-2015 [35] training sets, with results summarized in Tab. 1. On Sintel (train), OFM attains an EPE of 0.81 on the clean pass and 2.16 on the final pass, representing the strongest performance among all



Reference Frame I_1 | **OFM** | DPFlow | MemFlow

Figure 5. **Qualitative Comparisons on the Spring Test Set.** *Top:* predicted flow fields. *Bottom:* error visualizations. On the right, the colorbar for error visualization is shown, where red represents larger errors and blue indicates smaller errors (see the official website for more details). OFM delivers sharper motion boundaries and lower errors compared to DPFlow [36] and MemFlow [1].

Table 2. **Quantitative Results on Spring Benchmarks.** We compare OFM with state-of-the-art methods under zero-shot evaluation. Best and second-best results are shown in **bold** and <u>underline</u>, respectively. Multi-frame methods are marked in gray for reference.

Method	Venue	Spring (train)			Spring (test)			
		Extra data	1px ↓	EPE ↓	1px ↓	EPE ↓	Fl ↓	WAUC ↑
RAFT [51]	ECCV'20	✗	4.788	0.448	6.790	1.476	3.198	90.920
GMA [18]	ICCV'21	✗	4.763	0.443	7.074	0.914	3.079	90.722
GMFlow [57]	CVPR'22	✗	29.49	0.930	10.355	0.945	2.952	82.337
Flowformer [13]	ECCV'22	✗	4.508	0.470	6.510	0.723	2.384	91.679
MS-RAFT+ [14]	IJCV'24	✗	<u>3.577</u>	<u>0.397</u>	5.724	0.643	2.189	92.888
SEA-RAFT (L) [54]	ECCV'24	✓	3.842	0.426	-	-	-	-
DPFlow [36]	CVPR'25	✗	-	-	<u>4.277</u>	<u>0.479</u>	<u>1.602</u>	<u>93.822</u>
FlowSeek (L) [38]	ICCV'25	✓	3.838	0.402	-	-	-	-
OFM (ours)	–	✓	**3.496**	**0.361**	**3.660**	**0.468**	**1.477**	**94.462**
StreamFlow [50]	NeurIPS'24	✗	-	-	5.215	0.606	1.856	93.253
MemFlow [6]	CVPR'24	✗	-	-	5.759	0.627	2.114	92.253
MEMFOF [1]	ICCV'25	✓	-	-	3.600	0.432	1.353	94.481

published two-frame methods. On KITTI (train), it further sets new records with an EPE of 3.32 and an F1-all score of 10.9%, outperforming all recently reported approaches. For Sintel online testing, OFM achieves an EPE of 0.94 and 1.85, surpassing recent methods such as SEA-RAFT (L) [54] and DPFlow [36] by 28.5% and 8.4%, respectively. On KITTI online testing, it ranks second with a Fl-all score of 3.78, slightly behind DPFlow [36].

Qualitative comparisons with state-of-the-art methods are shown in Fig. 4. OFM recovers sharper boundaries and cleaner structure in regions with large motion or repetitive textures, where competing models often exhibit bleeding or over-smoothing. Its explicit motion-distribution formulation also yields more stable estimates in occluded areas, producing flows that remain coherent across both spatial detail and temporal evolution. Please refer to the official website for more comprehensive comparisons.

4.3. Results on Spring

Following most of the recent works, we report the results of the model trained on "C+T+S+K+H". The models are not fine-tuned on Spring data and are evaluated in a zero-shot setting. As presented in Tab. 2, on Spring (train), our approach achieves 1px and EPE scores of 3.496 and 0.361, surpassing the recent two-frame method FlowSeek (L) [38] by 8.9% and 10.2%, respectively. For online testing, OFM achieves the best performance on the official benchmarks, boosting the four standard metrics 1px, EPE, Fl, and WAUC to new records of 3.660, 0.468, 1.477, and 94.462, respectively. Moreover, even when compared with recent multi-frame approaches, OFM exhibits competitive or even superior performance. It significantly surpasses StreamFlow [50] and MemFlow [6] by 18.6% and 23.6% on average across all metrics. Qualitative examples from the Spring online test are shown in Fig. 5.

4.4. Ablation Study

OFM Formulations. To begin with, we compare different OFM formulations in Tab. 3. The OFM-baseline denotes the underlying optical flow backbone without temporal modeling or Flow Matching components. OFM-Naive, which directly applies the theoretical formulation, fails to

Table 3. **Ablation Study.** The default settings are marked in **bold**. 1-NFE means a single function evaluation, *i.e.*, one-step sampling.

Method	Setting	Sintel (train)		KITTI (train)	
		Clean	Final	EPE	Fl-all
Formulations	OFM-baseline	0.96	2.45	4.01	13.8
	OFM-Naive (1-NFE)	15.73	15.67	35.35	77.3
	OFM-Model (1-NFE)	1.49	2.88	7.14	17.0
	OFM-TVS (1-NFE)	0.84	2.25	3.85	12.5
	OFM-TVS (3-NFE)	0.82	2.19	3.53	11.7
Extra data	No	0.82	2.19	3.53	11.7
	TartanAir	0.82	2.16	3.32	10.9
Scale factor α	5	0.85	2.23	3.47	11.5
	10	0.82	2.16	3.32	10.9
	15	0.83	2.18	3.41	11.3
NFE steps K	1	0.85	2.22	3.61	11.6
	2	0.83	2.18	3.43	11.1
	3	0.82	2.16	3.32	10.9
	4	0.83	2.16	3.30	10.8
// All models are trained following their default setting (w/o extra data).					
Compatibility	RAFT	1.44	2.71	5.04	17.4
	RAFT + OFM	1.23	2.59	4.43	15.8
	SKFlow	1.22	2.46	4.27	15.5
	SKFlow + OFM	1.16	2.35	3.93	14.6

converge, supporting our observation in Sec. 3 that the raw objective is unstable when trained end-to-end. OFM-Model adds temporal embeddings but remains purely discriminative, and its performance drops below the baseline, indicating that temporal cues alone do not perform well. In contrast, OFM-TVS, *i.e.,* the full OFM configuration, consistently improves performance across all metrics. These results show that OFM is effective only when the architectural design, Flow Matching settings, and TVS strategy are used together, and neither component works in isolation.

Extra Data. Following the widely adopted TartanAir [53] rigid-flow pre-training protocol used in recent works [1, 54], we further evaluate OFM under this setting, as reported in Tab. 3. OFM already delivers strong results without TartanAir, and the rigid-flow initialization provides additional gains, particularly on KITTI, highlighting the model's ability to benefit from extra geometric supervision while remaining competitive even in its absence.

Scale Factor. Despite its strong theoretical foundation, our method remains exceptionally simple to implement, requiring the tuning of only a single hyperparameter, *i.e.*, the scale factor α. As shown in Tab. 3, we find that OFM is highly robust to a wide range of α values, indicating that the method's effectiveness does not hinge on precise tuning. For our final implementation, we set $\alpha = 10$, which offers a slight but consistent performance advantage over other tested configurations.

Sampling Steps. Consistent with Flow Matching–based generative models, OFM benefits from increasing the number of function evaluations (NFE) during inference. As shown in Tab. 3, OFM already delivers strong results with a single sampling step, surpassing most published two-frame methods. Increasing the number of steps from $K = 1$ to $K = 4$ yields further gains, reaching new state-of-the-art performance. To balance accuracy and computational efficiency, we adopt $K = 3$ as the default configuration.

Table 4. **Parameter and Runtime Comparison.** The reported time is for processing KITTI images with size of 376×1248.

Method	FlowFormer++ [42]	FlowDiffuser [31]	OFM (3-NFE)
Param. (M) $\downarrow$	16.2	16.3	15.6
Time (ms) $\downarrow$	375	260	270

Compatibility. To evaluate the compatibility of the proposed approach, we implement our OFM on recent renowned RAFT-style models, including RAFT [51] and SKFlow [49]. As shown in the last part of Tab. 3, OFM consistently boosts the performance of the base models by a clear margin. Note that we do not carefully tune the implementation details of the base models but simply apply our algorithm and the necessary time embedding modules. This demonstrates the strong compatibility of our approach.

Parameter and Runtime. Tab. 4 presents the comparison with recent approaches. Notably, the NFE steps of our OFM correspond only to part of the decoding process. Therefore, the increase in runtime is partial rather than multiplicative.

5. Conclusion

We presented Optical Flow Matching (OFM), a formulation that reframes optical flow estimation as continuous transport rather than discrete displacement. Besides, the proposed Triangle Velocities Synergy strategy aligns theoretical marginal velocities with usable training signals, enabling stable and efficient learning. Experiments on benchmarks show that OFM achieves state-of-the-art accuracy and superior temporal stability. We believe this continuous-transport view offers a promising direction for unifying motion estimation and dynamic modeling in future work.

Limitation and Future Work. Our OFM-TVS framework, grounded in conditional optimal transport and an Euler-based ODE solver, represents a classical solution but not an optimal one. Meanwhile, the Flow Matching field has progressed with more efficient methods such as MeanFlow [10] and Shortcut Models [9]. Our future work will focus on leveraging insights from these methods to improve OFM.

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